

OpenStax Chemistry 2e Errata

<https://openstax.org/details/books/chemistry-2e>

May 12, 2026 – Dean Peterson

Notes:

- I am a chemistry learner, not an expert, doing this avocationally not in a class setting.
- Since I lack instructor access, I could not review even numbered exercise solutions.
- I have attempted to include screenshots of unlabeled figures to assist in locating them.
- This document should have a PDF table of contents created from the headings.

Chapter 5

Chapter 5.1

Figure 5.4 alt text

The alt text refers to labels reading “Hot liquid.” and “Cold liquid.” However, on the figure the labels actually read “Hot water.” and “Cold water.”

Figure 5.5 alt text for part (a)

The alt text sentences 3 through 5 read:

The thermometer has a printed scale to the left of the tube in the center that reads from negative forty degrees at the bottom to forty degrees at the top. It also has a scale printed to the right of the tube that reads from negative thirty degrees at the bottom to thirty five degrees at the top. On both scales, the volume of the alcohol in the tube reads between nine and ten degrees.

However, it is not really two scales, it is one scale marked on both sides, with even multiples of 10 labeled on the left, and odd multiples of 10 labeled on the right. It is not Fahrenheit and Celcius, because the tick marks on both sides have the same separation. Therefore, I suggest replacing sentences 3 through 5 with

The thermometer has a tube in the center and a printed scale that reads from negative forty degrees at the bottom to forty degrees at the top. The volume of the alcohol in the tube reads between nine and ten degrees.

Even if you prefer to refer to them as separate scales, then sentence 4 still needs to be corrected. The right side does not read from -30 to +35. The right side reads from -40 to +40, just like the left side.

Link to Learning just below Figure 5.5

The link labelled “demonstration” is to the entire wikipedia page about Bimetallic strips. Is the intent to refer to the animated gif image on the page with the lighter? If so, would it make sense to link directly to that animated gif on Wikimedia commons instead?

https://commons.wikimedia.org/wiki/File:Bimetal_coil_reacts_to_lighter.gif

Figure 5.7 alt text for part (a)

Sentence 2 of the alt text reads (emphasis mine):

Picture a shows a metal railroad **tie** being cut with the flame of an acetylene torch. However, the metal piece being cut is not the tie. The tie is the wood holding the rails together. The metal is the rail, so sentence 2 should read (change in bold):

Picture a shows a metal railroad **rail** being cut with the flame of an acetylene torch.

Figure 5.7 alt text for part (b)

Sentence 3 of the alt text reads:

Picture b shows a chemical cold pack containing ammonium nitrate. However, the picture has both “ammonium nitrate” and an “Inner bag of water” labeled. Perhaps a more complete sentence 3 be (change in bold):

Picture b shows a chemical cold pack containing ammonium nitrate **and an inner bag of water**.

An alternative would be to reuse the more detailed description from the Figure 5.16 alt text for part (b), but that may be too detailed for the context. If you decide to reuse the Figure 5.16 alt text, first see the correction to the Figure 5.16 alt text given in a separate errata later in this document.

Chapter 5.2

Body text 1st paragraph below Figure 5.13, 2nd-to-last sentence

The 2nd-to-last sentence ends (emphasis mine):

... with no heat gained or lost by **either** its external environment.

The word “either” should be removed as there is only one thing, not two, after it.

Possible replacements for the end of that sentence include:

- ... with no heat gained or lost by its external environment.
- ... with no heat **either** gained or lost by its external environment.
- ... with heat **neither** gained **nor** lost by its external environment.

Figure 5.14 alt text clarity

The alt text is correct but an extra sentence might clarify that the thermometers are in the water but not touching the metal, something that is obvious visually. Sentence 4 reads:

Both containers are full of water, which is depicted by the letter “W,” and each container has a square in the middle which represents a metal which is labeled with a letter “M.”

Would it make sense to insert a new sentence 5 after sentence 4 reading something like:

The thermometer in each container is in the water, not touching the metal.

Example 5.3, first Check Your Learning solution

Why is the answer of 335.6 °C to 4 significant digits when many of the inputs are only given to 3 significant digits? Shouldn't it be 336 °C?

Example 5.4, last sentence of Solution section

The sentence reads:

Comparing this with values in Table 5.1, our experimental specific heat is closest to the value for copper (0.39 J/g °C), so we identify the metal as copper.

However, the value for copper in Table 5.1 is 0.385 J/g °C. Even if you wanted to round it to 2 significant digits, wouldn't it round down to 0.38 J/g °C, since the “8” is even and the “5” has no following digits?

Example 5.5, Check Your Learning section

The Check Your Learning section uses different rounding behavior than the Solution section above it. The Example 5.5 Solution section explicitly uses 2 significant figures for the density of water input to the calculation, saying “The density of water is approximately 1.0 g/mL, so 100.0 mL has a mass of about 1.0×10^2 g (two significant figures).” Note that the “(two significant figures)” is an original part of the text, not a clarification of my own. The Example 5.5 Solution gives its answer of 2.9 kJ to two significant figures. Therefore, I do not understand why the book gives the answer of Example 5.5 Check Your Learning with three significant figures as 1.34 kJ, instead of using 2 significant figures like the Solution section above it, which would be 1.3 kJ.

Figure 5.16 alt text

Sentence 4 reads (emphasis mine):

It also has three pictograms, which from **right to left**, show a hand squeezing the pack, agitating the pack and placing the pack on a person's body.

The phrase “right to left” should be “left to right”.

Figure 5.17 alt text

Sentence 5 reads:

Another container, labeled “bomb,” sits inside of a smaller cylinder which holds a sample cup and is nested in the cylindrical container surrounded by the water.

I think a better sentence 5 would read:

Submerged in the water is another, smaller cylinder labeled “bomb,” which holds a sample cup.

Reasoning follows:

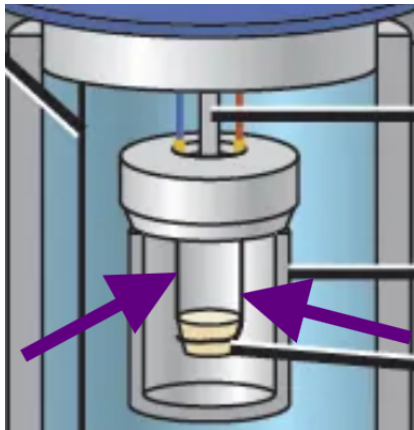
Following the original sentence 5 in detail seems to overstate the number of cylinders/containers by two and misstates which cylinder holds the sample cup. According to sentence 5 as written we have:

- A container labeled “bomb”, which is “inside of”
- A “smaller cylinder”, “which holds a”
- Sample cup, which is written as being inside the “smaller cylinder”, not the bomb,
- The “smaller cylinder” is nested inside of “the cylindrical container surrounded by the water.” (Note the “surrounded by”, not “containing” the water.)
- The outermost cylinder containing the water (from sentence 4, only implied in sentence 5.)

So that is a bomb cylinder inside 3 other cylinders and the sample cup not in the bomb. When I believe we actually have.

- A sample cup inside a
- Cylinder labelled “bomb”, submerged in
- The water, which is contained in
- An outer cylinder.

In the screenshot below, I added big purple arrows pointing to two vertical lines leading down to the sample cup. I believe the two vertical lines are just wires holding the sample cup. Perhaps the original alt text author interpreted those lines as the outline of a third nested cylinder?



Example 5.7 Solution section, final sentence

The final sentence reads:

This reaction released 48.7 kJ of heat when 3.12 g of glucose was burned.

According to the math section immediately above it, the 48.7 kJ should be 48.8 kJ instead.

$$\begin{aligned} q_{\text{rxn}} &= -(q_{\text{water}} + q_{\text{bomb}}) \\ &= -[(4.184 \text{ J/g } ^\circ\text{C}) \times (775 \text{ g})] \\ &= -(38,300 \text{ J} + 10,500 \text{ J}) \\ &= -48,800 \text{ J} = -48.8 \text{ kJ} \end{aligned}$$

This reaction released 48.7 kJ of heat

Figure 5.18 alt text

Sentence 8 ends (emphasis mine) ‘...above the words “Calories 250” and “Calories from fat 210.”’ The Calories from fat on the image are 110, not 210.

Link to Learning at bottom of Ch 5.2

The link is effectively broken. It takes you to a page. However, that page is a Dietary Reference Intakes calculator, not a nutrient database as described in the Link to Learning.

Chapter 5.3

Figure 5.19 alt text

What the alt text says is correct. However, I think the way the alt text is worded and ordered omit a spatial relationship on the diagram that may be meaningful. Specifically, the wording and order obscures the fact that the in-facing q_{in} and w_{on} arrows point to the half of the system oval containing the $\Delta U > 0$ relation, and the out-facing q_{out} and w_{by} arrows come from the half of the system oval containing the $\Delta U < 0$ relation. I could be wrong as I am a layperson learning chemistry. However, reading the body text, I think those spatial choices are significant. Someone may want to review the alt text to see if the spatial relationships between the arrows and the greater than and less than relations should be made more explicit. Thank you.

Example 5.9 Check Your Learning

The book gives the answer as $\Delta H = -338$ kJ. I don't know if graders would mark this down, but I believe the actual answer is $\Delta H = -339$ kJ. I tried different atomic weight sources and adding intermediate rounding, but was unable to reproduce the book's answer of -338 that way. The only way I could reproduce the book's answer of -338 was to both A) ignore that the grams of the product is given so I can calculate directly from that instead of worrying about which reactant is the limiting reactant, and B) mistakenly determine iron instead of chlorine as the limiting reactant and do the calculation starting from the grams of iron.

The procedure I followed to get $\Delta H = -339$ kJ:

1. Notice that, unlike the Example 5.9 Solution section, the question is asked in terms of the $\text{FeCl}_2(\text{s})$ product not one of the reactants, and that the grams of $\text{FeCl}_2(\text{s})$ are given, eliminating the need to determine the limiting reactant.
2. Calculate the molecular mass of FeCl_2 as:
 $55.85 + 2 * 35.45 = 126.75$ amu.
3. Calculate the moles of FeCl_2 in 3.22 g of FeCl_2 as:
 $3.22 / 126.75 = 0.0254043$ mol.
4. Calculate the heat that would be produced in creating 1 mole of FeCl_2 as:
 8.60 kJ (for 3.22g) / 0.0254043 (mol in 3.22g) = 338.525 kJ
5. Round and reverse the sign to give $\Delta H = -339$ kJ.

Even when I tried using the amounts of the reactants, I found that 1.42 g of Fe is 0.025425 mol and that 1.80 g is 0.025388 mol of diatomic Cl_2 . Meaning that Cl_2 , not Fe, is the limiting reactant (or that neither was the limiting reactant and it was within the measurement error I suppose). Calculating ΔH based on the amounts of Fe and Cl_2 gives -338.744 kJ calculated from the limiting reactant Cl_2 , which again rounds to -339 kJ. Only if you calculate from the non-limiting reactant Fe do you get -338.246 kJ which again rounds to the book answer of -338 kJ.

Figure 5.21 alt text

From the picture you cannot really tell whether the firefighter is male or female. Would it be slightly more gender neutral, and more accurate to the limitations of our knowledge, to replace the word "fireman" in the alt text with "firefighter"?

Example 5.10 Solution, value used for isooctane enthalpy of combustion

The solution text says (emphasis mine) “The enthalpy of combustion ... Table 5.2 gives this value as **-5460** kJ per 1 mole of isooctane (C₈H₁₈).” The solution equation also uses the -5460 kJ/mol value. However, Table 5.2 actually gives the enthalpy of combustion for isooctane as **-5465.5** kJ/mol. Due to rounding, this does not seem to change the final answer of 33,100 kJ.

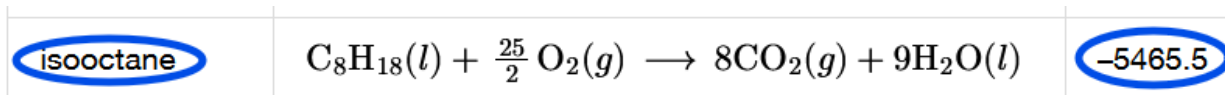


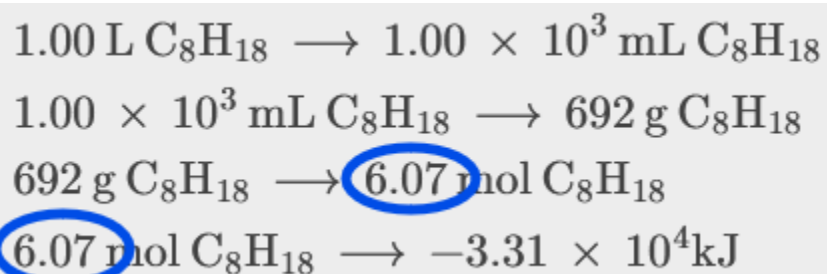
Table 5.2

Table 5.2 gives this value as -5460 kJ per 1 mole of isooctane

$$\frac{0.692 \text{ g C}_8\text{H}_{18}}{1 \text{ mL C}_8\text{H}_{18}} \times \frac{1 \text{ mol C}_8\text{H}_{18}}{114 \text{ g C}_8\text{H}_{18}} \times \frac{-5460 \text{ kJ}}{1 \text{ mol C}_8\text{H}_{18}}$$

Example 5.10 Solution, intermediate number of moles

After giving the final answer, the solution lists intermediate results in a sequence of steps. I believe the 6.07 mol C₈H₁₈ intermediate answer is wrong. I do not know if an alternate atomic weight source was used or what. Using the atomic weights in the Appendix A, (12.01 × 8) + (1.008 × 18) = 114.224. 692 g / 114.224 = 6.05827146659196, which rounds to 6.06, not 6.07.



Details which might not matter:

Note when using the incorrect enthalpy of combustion for isooctane of -5460 (previous issue), this 6.07 intermediate result does not affect the final rounded result of 33,100 kJ. However, if you use the corrected Table 5.2 value for the enthalpy of combustion for isooctane, -5465.5, multiplying by the bad intermediate result of 6.07 moles rounds to 33,200 kJ, not 33,100 kJ. If both issues are corrected, though, the final result remains 33,100 kJ. That was probably clear as mud, so...

Table of details which might not matter:

Description	mol C ₈ H ₁₈	kJ/mol C ₈ H ₁₈	kJ (unrounded)	kJ (rounded)
Neither issue corrected	6.07	-5460	-33,142.2	-33,100
Enthalpy corrected	6.07	-5465.5	-33,175.585	-33,200
Intermediate result corrected	6.06	-5460	-33,087.6	-33,100
Both issues corrected	6.06	-5465.5	-33,120.93	-33,100

Figure 5.22 (b) alt text

Sentence 4 reads (emphasis mine):

Picture b shows five **large tubs** full of a brown liquid containing these algal organisms. If you look closely, picture b appears to be an aerial photograph and the lines between the “tubs” in the background are roads. Therefore, I wonder if the words “large tubs” should be replaced with “**ponds**” to more accurately reflect scale.

Figure 5.23 alt text

This doesn't matter much, but sentence 4 starts (emphasis mine):

Above each term, respectively, are diagrams of three containers, **three** cylinders lying side-by-side, ...

The diagram actually shows **four** cylinders, not three.

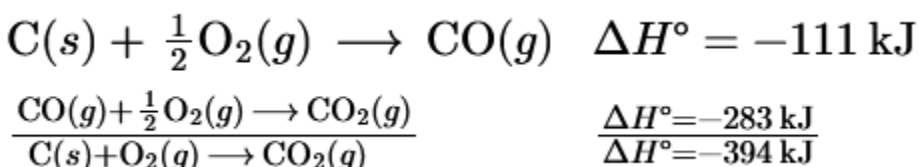
Standard Enthalpy of Formation Section, paragraph before Example 5.11

Suggestion: Appendix G is introduced in this paragraph. Would it be worth mentioning that students can look up the standard stable form of elements by finding which form has 0 for the standard enthalpy of formation column in Appendix G? (At least I think that's the case.) It could go at the end of this paragraph or as part of working an example like Example 5.12 where the student needs to know that P₄(s) is the standard state of phosphorus.

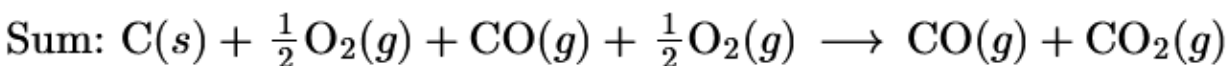
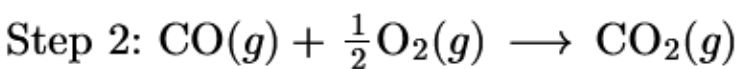
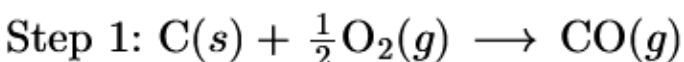
Hess's Law section, 6th display equation formatting

The sizing of the bottom two lines of the 6th display equation is odd, smaller than the top line. When combined with the horizontal rules for the summation, it makes the bottom two lines almost look like fractions rather than the continuation of the summation from the top line. The

following screenshots are from Firefox browser on Windows 11, but Chrome and Edge browsers on Windows 11 show the same issue. Chrome browser on an Android tablet also shows the issue.



Compare this with the earlier 4th display equation in the Hess's Law section, where the even sizing gives the correct visual impression that this is a summation.



In addition to dealing with the sizing, it might not be a bad idea to add the “Step 1”, “Step 2”, and “Sum” labels to the beginning of the lines in the 6th display equation just like are seen in the 4th display equation.

Note: Similar issues exist in Examples 5.13 and 5.14 and are described in their own errata below.

Figure 5.24 alt text, numeric values

The alt text contains the values “Δ H = −394 k J”, and “Δ H = −283 k J”. The visual diagram has the slightly more precise values of, “Δ H = −393.5 k J”, “Δ H = −282.5 k J”. Note that while the alt text values do not match the figure, they do match the values used in the equations in the body text section just above the figure. In other words, the values appear in three places, the body text formulas, the figure, and the figure alt text. The body text and alt text values match. The figure values are more precise.

Note: The suggested fix for the following, separate Figure 5.24 alt text errata fixes the values to match the visual figure.

Figure 5.24 alt text, inaccurate descriptions of what labels point to

The last 7 sentences of the alt text read:

There are three brackets to the right of the diagram. The first bracket runs from the top horizontal line to the second horizontal line. It is labeled, “Enthalpy of reactants.” The second bracket runs from the second horizontal line to the bottom horizontal line. It is labeled, “Enthalpy of products.” Both of these brackets are included in the third bracket which runs from the top to the bottom of the diagram. It is labeled, “Enthalpy change of exothermic reaction in 1 or 2 steps.”

This attributes the label “Enthalpy of reactants” to the top bracket, “Enthalpy of products” to the bottom bracket, and “Enthalpy change of exothermic reaction in 1 or 2 steps” to the overall bracket. I believe these attributions are in error. I see this from three lines of evidence.

- Visually, the “Enthalpy of reactants” and “Enthalpy of products” labels have lines pointing not to the brackets, but to the top and bottom horizontal blue lines.
- The top and bottom brackets each span a reaction between two horizontal lines. A reaction has both reactants and products, so it would be incorrect to label the brackets as dealing with just reactants or just products. The brackets cover a change from one to the other.
- The top and bottom brackets cover vertical distance representing enthalpy change, not a single enthalpy value as those labels imply.
- The text of the third label, “Enthalpy change of exothermic reaction in 1 or 2 steps.” The big bracket is 1 step, the two smaller brackets are 2 steps. In other words, the two smaller brackets don’t need separate labels because that one label describes all the brackets together.

I think a more correct alt text for Figure 5.24 would read as follows:

A diagram is shown. A long arrow faces upward on the left with the phrase “H increasing.” A horizontal line at the top of the diagram has the formulas “C (s) + O subscript 2 (g)” above it. This line is labeled, “Enthalpy of reactants.” A horizontal line at the bottom of the diagram is shown with the formula “C O subscript 2 (g)” below it. This line is labeled, “Enthalpy of products.” The top and bottom lines are connected by a downward facing arrow with the value “ $\Delta H = -393.5 \text{ k J}$ ” written beside it. Below and to the right of the top horizontal line is a middle horizontal line with the equations “C O (g) + one half O subscript 2 (g)” above it. This middle line and the top line are connected by a downward facing arrow with the value “ $\Delta H = -111 \text{ k J}$ ” written beside it. The middle line and the bottom line are connected by a downward facing arrow with the value “ $\Delta H = -282.5 \text{ k J}$ ” written beside it. There are three brackets to the right of the diagram. The first bracket runs from the top horizontal line to the middle horizontal line. The second bracket runs from the middle horizontal line to the bottom horizontal line. Both of these brackets are included in the third bracket which runs from the top to the bottom of the diagram. To the right of the brackets is a label, “Enthalpy change of exothermic reaction in 1 or 2 steps.”

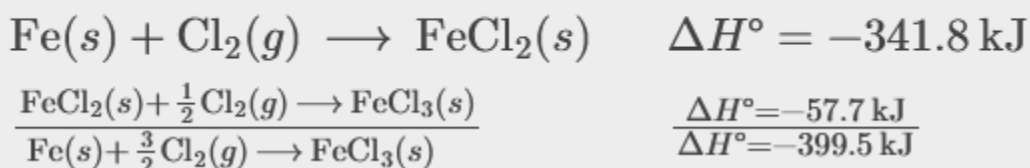
That alt text has the following changes:

- It fixes this label attribution issue.
- It fixes the incorrect (compared with the diagram) kJ value issue described in the other figure 5.24 alt text errata above. If the decision for that errata is to change the visual figure not the alt text, then those changes should not be made to the alt text.
- It reorders the descriptions from top to bottom, from reactants to products.
- It replaces the term “second horizontal line” with “middle horizontal line” because it seemed clearer to me, especially since the middle horizontal line is the third horizontal line mentioned, not the second.

Example 5.13 Solution, 2nd display equation formatting

This equation has the same format sizing issue as the Hess's Law section 6th display equation issue described above.

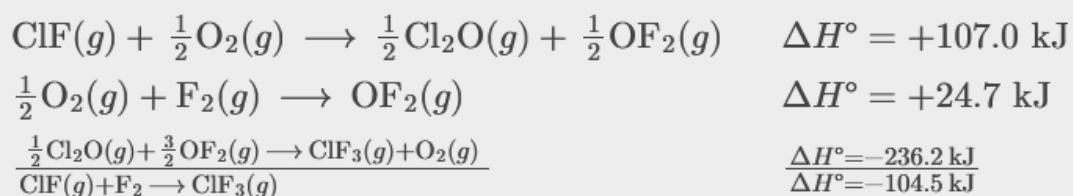
known ΔH values, so we must sum their ΔH s:



Example 5.14 Solution, 4th display equation formatting

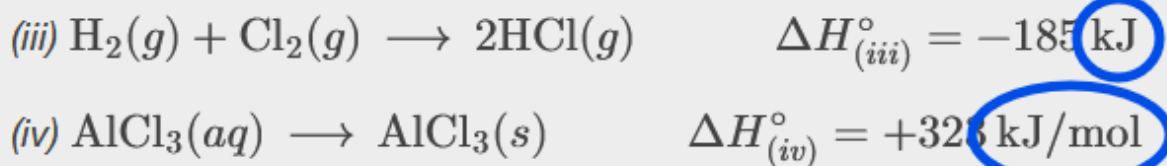
This equation has the same format sizing issue as the Hess's Law section and Example 5.13 issues described above.

Now check to make sure that these reactions add up to the reaction we want:



Example 5.14, Check Your Learning section

Equation (iv) gives its ΔH° in units of kJ/mol, whereas equations (ii), (iii), and (v) give their ΔH° in units of kJ. Is this difference intentional?



Example 5.15, section Solution: Using the Equation

Since two of the inputs, the enthalpies of formation of HNO_3 and NO_2 , are only given to 1 digit after the decimal point, shouldn't the answer be rounded to 1 digit after the decimal point. That is, shouldn't the answer be -138.3 kJ instead of -138.32 kJ? I am basing this on the rounding rules in Chapter 1.5, section Significant Figures in Calculations, rule 1.

Example 5.15, the two Solution sections do rounding differently

The section Solution: Using the Equation does not round and gets an answer of -138.32 kJ.

$$\begin{aligned} &= [2 \times (-206.64) + 90. \\ &= -323.03 + 186.23 \\ &= -138.32 \text{ kJ} \end{aligned}$$

However, the section Solution: Supporting Why the General Equation Is Valid performs the same calculation by another means and gets a different answer of -138.4 kJ.

Summing their enthalpy changes gives the value we want to determine:

$$\begin{aligned} \Delta H_{\text{rxn}}^{\circ} &= \Delta H_1^{\circ} + \Delta H_2^{\circ} + \Delta H_3^{\circ} + \Delta H_4^{\circ} = (-99.6 \text{ kJ}) + (+285.8 \text{ kJ}) \\ &= -138.4 \text{ kJ} \end{aligned}$$

So the standard enthalpy change for this reaction is $\Delta H^{\circ} = -138.4 \text{ kJ}$.

The second solution section rounds at each intermediate step, which appears to explain the discrepancy. The principle is still demonstrated, so it may not matter, but I thought I would point out the discrepancy.

(Out of order) Chapter 1.5, section Significant Figures in Calculations

Suggestion: Apologies if the book contains an example emphasizing this and I just missed it. However, would it make sense to include a final Example in Chapter 1.5 that illustrates how the significant figure rule for multiplication and division only applies to measurements, not precise counts. I am thinking of how the multiplications in the Example 5.15 Solution Using the Equation section are multiplying by stoichiometric coefficients, not measurements, and therefore the significant figure rules for multiplication do not apply. The book's body text does mention this, both at the start of Chapter 1.5 where it says (emphasis mine), "Quantities derived from measurements **other than counting**, however, are uncertain..." It also implies this at the start of the Chapter 1.5 section Significant Figures in Calculations section when it says (emphasis mine) "...results **calculated from a measurement** are at least as uncertain as the measurement itself." Despite the body text mentioning it, I don't think that distinction is shown in an example. I was thinking that an example where some operations involve uncertain measurements and some operations involve exact counts might help clarify the issue in a student's mind. Just a thought.

Chapter 5 - Exercises

Note: Only odd exercises reviewed because I do not have instructor access.

Exercise 5.5 (a) rounding

The book answer is $47.6 \text{ J/}^\circ\text{C}$; $11.38 \text{ cal } ^\circ\text{C}^{-1}$. Why is the calorie answer expressed to 4 significant figures when the input of 45.8 g N_2 is given to 3 significant figures. Shouldn't it be $11.4 \text{ cal } ^\circ\text{C}^{-1}$?

Exercise 5.5 (b) answer

The book answer is $407 \text{ J/}^\circ\text{C}$; $97.3 \text{ cal } ^\circ\text{C}^{-1}$. Shouldn't the calorie answer be $97.2 \text{ cal } ^\circ\text{C}^{-1}$? The weight of Aluminum in the problem is given as 1.00 pounds. The specific heat of Al is given in Table 5.1 as $0.897 \text{ J/g } ^\circ\text{C}$. The number of joules per calorie is given in both the Chapter 5.1 text and in Appendix C as 4.184 joules/cal . I tried two different metric conversions from Appendix C. Neither gave me an answer that rounded to 97.3. Both gave me 97.2 after rounding.

Source	Pound conversion	Calculation	Result
Appendix C	$1 \text{ lb} = 0.4535924 \text{ kg}$	$1 \times 0.4535924 \times 1000 \times 0.897 / 4.184$	97.24483
Appendix C	$1 \text{ kg} = 2.205 \text{ lb}$	$1 / 2.205 \times 1000 \times 0.897 / 4.184$	97.22818

Exercise 5.13 numeric answer and rounding

The book answer for energy required is 7.47 kWh . Shouldn't the answer be rounded to 2 significant figures as 7.5 kWh ? The following inputs are all 2 significant figures:

- The initial and final temperatures, 85 and $72 \text{ }^\circ\text{F}$.
- The Joules to kWh conversion factor 3.6×10^6 .
- The assumed density of water 1.0 g/mL .

If the answer should be three significant figures, I think it should be 7.49 kWh , not 7.47 kWh . At least this is true if you do the entire calculation in one step with no intermediate rounding, as recommended in Chapter 3.3 in the body text just after Example 3.17. For example, plugging the following calculation into Python or even Google search gives 7.48729 , which rounds to 7.49 .

$$-(((72 - 32) / 9 * 5) - ((85 - 32) / 9 * 5)) * 892 * 10^{**3} * 1.0 * 4.184 / (3.6 * 10^{**6})$$

Exercise 5.13 text answer

The last sentence of the problem asks of assumptions "How did they affect your calculated result (i.e., were they likely to yield "positive" or "negative" errors)?" However, the Exercise 5.13

solution in the book only lists assumptions, not the likely direction of impact as asked in the final sentence.

Exercise 5.29, issue 1

The exercise asks "...how much would the temperature change?". However, the solution given, 22.6, is not the temperature change, but the final temperature. The clearest way to resolve this might be to clarify the question. Instead of asking for the temperature change, ask for the new final temperature.

Exercise 5.29, issue 2

Regardless of how issue 1 is resolved, there is an additional issue where a more careful student could get the wrong answer. I see at least three ways of working the problem:

- Use the procedure outlined in the explanation portion of the Exercise 5.29 solution where it says "...the two-fold increase in the amount of water leads to a two-fold decrease of the temperature change." That would lead to an answer of:
 $24.9 - ((24.9 - 20.3) / 2) = \mathbf{22.6\text{ }^{\circ}\text{C}}$
- Work backward from the **rounded** final answer of heat absorbed in Example 5.6, 1.0 kJ, taking into account the mass of the solute, to get:
 $24.9 - (1000 / ((100.0 + 3.21) * 4.184)) = 22.584\text{ }^{\circ}\text{C}$, which rounds to **22.6 °C**.
- Work backward from **unrounded** calculations of heat absorbed in Example 5.6 to get:
 $q_{5.6} = -(50.0 + 3.21) * 4.184 * (20.3 - 24.9) = 1024.1\text{ J}$, then
 $24.9 - (1024.1 / ((100.0 + 3.21) * 4.184)) = 22.528\text{ }^{\circ}\text{C}$, which rounds to **22.5 °C**.

Note how the most careful route gives a different answer. I would hate to see a student marked wrong for being extra careful. Would it be giving too much of a hint to explicitly say the Joules of heat absorbed that you want the student to use, such as 1.0 kJ, in the question of Exercise 5.29?

Note: The solutions to exercises 5.41 and 5.43 have similar issues.

Exercise 5.29, issue 3

The numeric part of the solution, 22.6, is given without units, unlike other answers.

Exercise 5.31 solution

I was only able to get the book solution of 11.7 kJ if I did intermediate rounding. If you avoid intermediate rounding, and plug the following into python or Google search you get:

$$(534 + (675 * 1.0 * 4.184)) * (26.9 - 23.4) = 11753.7\text{ J, which rounds to 11.8 kJ.}$$

Exercise 5.41 solution

This exercise has an issue similar to Exercise 5.29 issue 2. If you base the exercise calculation off of the rounded final value for Example 5.6, $q_{\text{rxn}} = +1.0 \text{ kJ}$, you get the book answer of 25 kJ mol^{-1} . However, if you base the exercise calculation off of the unrounded value for Example 5.6, $q_{\text{rxn}} = 1024.1 \text{ J mol}^{-1}$, then you get a different answer of 26 kJ mol^{-1} . I am basing that off of this python calculation, which can also be pasted into a Google search box:

$$1024.1 * (14.01 + 1.008 * 4 + 14.01 + 16.00 * 3) / 3.21 = 25539.33 \text{ J mol}^{-1}.$$

Again, I would hate for a more careful student to get marked down. Would it be giving too much of a hint to explicitly say the Joules of heat absorbed that you want the student to use, such as 1.0 kJ , in the question of Exercise 5.41?

Exercise 5.43 solution, issue 1

The book gives an answer of 81 kJ mol^{-1} (positive). Since the answer to Exercise 5.25 is -2.2 kJ , and since the reaction is exothermic, shouldn't the sign be negative, as in -81 kJ mol^{-1} ?

Exercise 5.43 solution, issue 2

This exercise also has an issue similar to Exercise 5.29 issue 2 and Exercise 5.41. If you base the exercise 5.43 calculation off of the rounded answer for exercise 5.25, -2.2 kJ , you get the book answer magnitude of 81 kJ mol^{-1} . However, if you base the exercise 5.43 calculation off of the unrounded answer to exercise 5.25, -2174.436 J , you get a different answer magnitude, 80 kJ mol^{-1} .

Exercise 5.45 solution

The book answer is 5204.4 kJ . However, since one of the inputs, 4.00 moles , is only given to three significant figures, shouldn't the answer be rounded to three significant figures, and therefore be 5200 kJ , or $5.20 \times 10^3 \text{ kJ}$?

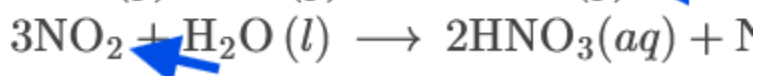
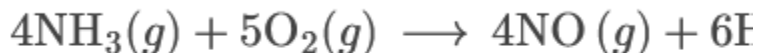
Exercise 5.57 solution

This might not matter, but the only inputs are -1378.83 kJ/mol from Appendix G, to 6 significant figures, and exactly 1 mole. Therefore, technically shouldn't the answer be kept to 6 significant figures as well, as 459.610 kJ , instead of the book answer of 459.6 kJ ?

Exercise 5.59 question

In the third chemical formula, reactants side, the 3NO_2 term appears to be missing the state of matter (g).

59. The following sequence of reactions occur



Exercise 5.67 solution

The book gives the answer as 90.3 kJ/mol. However, the exact answer is:

$$(66.4 + 114.1) / 2 = 90.25 \text{ kJ/mol}$$

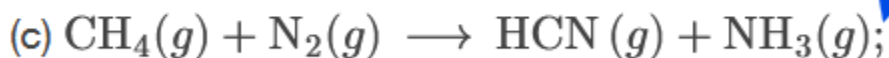
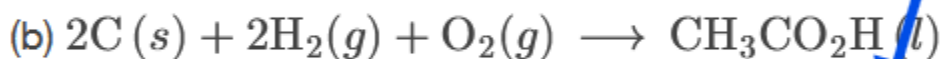
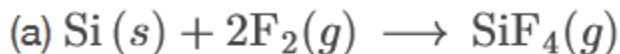
According to the rounding rules in Chapter 1.5, section Significant Figures in Calculations, "If the dropped digit is 5, and it's either the last digit in the number or it's followed only by zeros, round up or down, whichever yields an even value for the retained digit."

Therefore, I think the rounded answer should actually be 90.2 kJ/mol, not 90.3 kJ/mol.

Exercise 5.69 question

Ok. This is the smallest thing ever, and I only noticed it because I copied and pasted some stuff while working the problems. Why does only part (c) of the question end in a semicolon when none of the other parts do.

69. Using the data in [Appendix G](#), calculate the standard



Exercise 5.69 solution

It may not matter much, but why do the answers for parts (a) and (b) have units of kJ mol^{-1} , but the answers to parts (c) and (d) have units of just kJ? The question text is the same for all parts.

69. (a) $-1615.0 \text{ kJ mol}^{-1}$; (b) $-484.3 \text{ kJ mol}^{-1}$; (c) 164.2 kJ ; (d) -232.1 kJ

Exercise 5.71 solution

Shouldn't the sign of the answer be positive, not negative? The enthalpy change, ΔH , would be negative, but that means it is exothermic, right, and the question is asked as "...how much heat is produced..." This may require removing the per mole from the units to make sense I suppose. Therefore, I think the answer should be 54.04 kJ , not $-54.04 \text{ kJ mol}^{-1}$. (Also, considering physics, any reaction used to drive reaction control thrusters in space vehicles would need to cause expansion, which would seem to mean the reaction would have to give off heat, not absorb it.)

Exercise 5.73 solution

The book gives a solution of $-2660 \text{ kJ mol}^{-1}$. Shouldn't the final answer be given to 4 significant digits as $-2657 \text{ kJ mol}^{-1}$? I understand that the enthalpy of formation for butane is only given to 3 significant digits as -126 kJ/mol . However, the last step of the calculation is addition, where the significant digits in the answer should not be based on significant digit count like multiplication, but on rounding the result "to the same number of decimal places as the number with the least number of decimal places." In this case, that is the enthalpy of formation for butane, which includes the one's place, so I would think the answer should include the one's place as well. (Quote is from Chapter 1.5 section on Significant Figures in Calculations.)

Exercise 5.79 question

The first sentence of the question says (emphasis mine):

The enthalpy of combustion of hard coal averages -35 kJ/g , that of gasoline, **$1.28 \times 10^5 \text{ kJ/gal}$** .

Shouldn't the value for gasoline be negative, not positive? Shouldn't it be $-1.28 \times 10^5 \text{ kJ/gal}$? If you work the problem as given, you get a negative number of kilograms of coal for the answer, which is rather odd.

Exercise 5.81 question

The question begins (emphasis mine):

Among the substances that react with oxygen and that have been considered as potential rocket fuels are diborane [**B₂H₆**, produces B₂O₃(s) and H₂O(g)], methane [**CH₄**, produces CO₂(g) and H₂O(g)], and hydrazine [**N₂H₄**, produces N₂(g) and H₂O(g)]...

The products of the reactions have the states of matter like "(s)" appended. However, the reactants being combusted do not. This makes it ambiguous or at least harder to work the problem because one of the reactants, hydrazine [N₂H₄] has two entries in the table in Appendix G, one for N₂H₄(l) and one for N₂H₄(g). This does not affect the answer. If you work out the heat produced for both states of hydrazine, neither is competitive with the eventual answer diborane. However, it could cause confusion.

I recommend adding the states of matter to the **B₂H₆**, **CH₄**, and (especially) **N₂H₄** reactants in the question.

Exercise 5.85 (d) answer

Can you please double check the answer for part (d)? The book says 75.4 °C. I get 75.2 °C as follows. The math can be copied into python or Google search bar:

- Step 1: number of moles in 25 g of propane:
 $25.0 / (3 \times 12.01 + 8 \times 1.008) = 0.56697 \text{ mol}$
- Step 2: Temperature change for 4000 g of water, where -2219.2 is the heat of combustion of propane given in the question in part (c):
 $(-(-2219.2) \times 1000 \times 0.56697) / (4000 \times 4.184) = 75.18 \text{ °C}$ which rounds to 75.2 °C

Appendix C

Appendix C, section Units of Volume

Suggestion: In the Units of Volume row for the liter, one of the conversions given is for “(US) quarts”. However, subsequent rows have two different quarts, “liquid quart (US)” and “dry quart”. The liter row in the conversion table might be slightly clearer if its third conversion specified liquid quarts, such as = 1.057 (US) liquid quarts.

Units of Volume

liter (L)	= 0.001 m³ (exact, definition) = 1000 cm³ (exact, definition) = 1.057 (US) quarts
milliliter (mL)	= 0.001 L (exact, definition) = 1 cm³ (exact, definition)
microliter (μL)	= 10⁻⁶ L (exact, definition) = 10⁻³ cm³ (exact, definition)
liquid quart (US)	= 32 (US) liquid ounces (exact, definition) = 0.25 (US) gallon (exact, definition) = 0.9463 L
dry quart	= 1.1012 L
cubic foot (US)	= 28.316 L